

KANAK RAJ S

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Embedded Software Architect

Software Development | Continuous Improvement | Technical Expertise in Embedded Domain

Summary

18+ years' experience in embedded software development, consistency throughout all phases of software development projects. Currently working as Senior Software Project Dev Lead with Visteon, Chennai.

Software Engineering Skills

- Embedded Software Development
- Machine Product Development
- Device driver development
- Debugging
- Technical Leadership
- Global Project Integration
- Understanding of Microcontrollers
- Knowledge on RTOS
- CyberSecurity SW and Autosar
- TARA

Language/Tools/OS :

Embedded C , OSEK OS, Autosar Architecture(MCAL,BSW,Application), Dassault Autosar Builder, Davinci configurator, EB Tresos, Python ,canapé, CanAnalyser, Lauterbach Trace32 Debugger, GHS compiler, Jira, Jenkins, Doors, Git, Linux, Function generator, oscilloscope, MATLAB – Simulink, Stateflow, M-scripts,Jira.

Professional Experience

Visteon, Chennai | Senior Software Project Development Lead/Architect– Software | May 2025

- Sr SW Project lead for Porsche and Mercedes Benz- Battery Management system projects.
- Responsible for coordinating deliveries between multiple internal teams - Bootloader,Cyber security, Autosar , integration team, with clear understanding of technology.
- Responsible for resolving dependencies and roadblocks for the tech teams
- Responsible for high-level design (HLD) of the programs.
- Responsible for collaborating with software program delivery teams, architects and other teams to drive the development, testing and implementation of projects.
- Responsible for handling the external ASPICE process.
- Responsible for sprint planning and grooming stories/epics and Defect management.

VALEO, Chennai | Principal Engineer (Software Architect) – Software | Jan 2024 –April 2025

- Analyzed LiDAR Gen3 Software Requirement & Architecture with Real time Sensor data collection, aggregation, and interpretation.
- Collaborated with diverse Project Stakeholders to manage, analyze, and evaluate Software Requirements and functionality.
- Responsible for designing software architecture and adding software requirements in HLD(SWE2) based on the system/customer requirements.
- Facilitated Technical Reviews to implement Change Requests across Software Components, Design, and Code.
- Experience in Memory management, CPU load analysis, stack analysis(RAM &ROM analysis)
- Experience in handling all the ASPICE process and responsible for ensuring the software is approved by the quality team.
- Aided the Technical Project Manager in orchestrating Ticket and Defect management for test & development Strategy/Plan.

LTTS, Chennai | Senior Technical Specialist – Software | April 2023 –Nov 2023

Client - Marelli

- Development of CyberSecurity features for 12 inch cluster using vector sip package and Infineon Traveo controller.
- Responsible for configuring AUTOSAR security components – Csm,CryIf,Crypto,KeyM,CertM
- Responsible for developing Security features – Secure Boot Sequence (Chain of Trust), Secure Image validation, vHSM updater.
- Responsible for configuring HSM module and HOST modules.Also establish proper communication between them using IPC.
- Knowledge and configuration on cryptographic libraries.
- Experience in creating test plan, Unit test case creation and Integration testing.

LTTS, Chennai | Senior Technical Specialist – Software | Jan 2022 –March 2023

Client - Texas Instrument

- Experience in MCAL and IO modules - ADC, PWM, CAN, DIO, ICU, FLASH, SPI, CAN, CDD IPC for TI Microcontroller(SOC) device drivers.
- Responsible for AUTOSAR MCAL and Complex Driver software development, reviews and technical leadership to the team in the areas involving ISO26262 Functional safety.
- Experience in working on MCAL configuration Tool - EB Tresos.
- Analysis and porting of features from one controller to another.

- Experience in creating test plan, test creation and dynamic test analysis using code coverage tool- LDRA.Experience in using static code analysis using Klocworks tool.
- Experience in analysing Functional safety requirements, creating functional safety documents and implementing them.

Caterpillar India Private Ltd, Chennai | Senior Engineer – Software | June 2011 – Dec 2021

- Implementing various application features and releasing of final application software deliverables as flash file for cat machines like Hydraulic Mining shovel, Excavator, Large wheel loader.
- SME for different Software teams in CAT India involved in transitioning/Upgrading from the existing software architecture to Autosar architecture.
- Hands on experience on autosar tools like Autosar Builder, Davinci configurator and generate autosar complaint code.
- International deputation to **USA (a year and half)** and experience to interact with different teams and customers onsite.
- Interacting with various subsystems team, integrating application software with several new libraries and testing the final flash file on the ECM.
- Porting application software across multiple ECM platforms because of changes in technology.
- Experience on working with communication protocols like J1939 and CAN protocols to transfer data across different ECM's. Hands on experience on Can Analyser to test these communication protocols.
- Designing diagnostics strategies/algorithms to detect and raise diagnostics accurately and saving warranty cost for company, reduce machine downtime for customer.
- Understanding operator/dealer's (End customer) problems and design the diagnostics strategy accordingly
- Collate diagnostics issues on the customer machines on field and provide appropriate solutions for future machines
- Designed and implemented diagnostics detection and input processing algorithms in Simulink/state flow for different sensors like PWM, Active Analog/Passive Analog temperature and pressure sensors, Hall Effect sensors, Speed sensors, Switches.
- Designed and implemented diagnostics detection and output processing algorithms in Simulink/state flow for different solenoids like ON/OFF and Proportional solenoids.
- Experience in reading and comprehending the data sheet/sensor/solenoid specifications from different vendors.
- Autogenerate C code from Simulink/state flow models using RTW. Support different teams to integrate this autogenerated code with handwritten C code and resolve any issues.
- Hands on experience on tools like canape to measure/calibrate/override various data on the software.
- Hands on experience in using lab tools like Function generator, oscilloscope and Lauterbach-Trace32 debugging tool.

- Designed a software build process (using python) software builds in the night. This had an R&E efficiency of about \$20000 on each program executed.
- Automated many manual tasks by creating libraries using various tools like Matlab, python thereby saving time and improving efficiency.

Lear Automotive, Mumbai | Software Engineer | June 2010 - May 2011

- Development of body control Module features
- Understand the requirements and collaborate with the Product Team in Spain.
- Development of Visibility Functionality Models for Volvo and Ford cars using Simulink/stateflow - WindScreen cleaning, Headlamp Cleaning and Defroster Functionality for Front, Rear and Outer mirrors.
- Testing the code on the Target. Functional testing is done on the hardware panel.

Tata Consultancy Services, Pune | Assistant Associate Engineer | Mar 2007 - June 2010

- Development of power train features using Simulink/Stateflow models used in Nissan cars.
- Reverse Engineering - Involved in converting the existing C code to Simulink/Stateflow model for JATCO Motors.
 - Ensuring proper scaling and resolution were used.
 - Create test cases and perform MIL testing on the models. Generate and analyse the coverage report.
 - Created DLL's for the C code using Watcom compiler. These DLL's were interfaced with Simulink using S-function blocks and the results were compared model outputs.
- Developed customised additional guidelines using M-scripts and integrated them to the already existing MAAB guidelines of Simulink so that they can be run on the Simulink/Stateflow models using the Model Advisor option.
- Hands on experience on REACTIS tool to create test cases for model coverage.

Certifications

- ISO/SAE 21434:2021 Automotive Cybersecurity Engineer Professional Certification -Level 1 (Omnex India Pvt. Ltd.)

Education

B.Tech (Electronics & Communication -2006), JNT University, Hyderabad
